

# Jamie Seoh

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## EDUCATION

### University of Waterloo

*Bachelor of Software Engineering*

Sep 2025 – Present

3.95/4.00 (91.18%) GPA

- Logistics Organizer @Hack the North, Prev. Sponsorship Director @Ignition Hacks & Events @UWKSA

## TECHNICAL SKILLS

**Languages:** Python, C/C++, TypeScript & JavaScript, HTML & CSS, SQL, Java, R

**Frameworks and Libraries:** PyTorch, NumPy, React.js, React Native, Next.js, Svelte, FastAPI, Flask, Tailwind CSS

**Technologies:** Docker, GCP, Azure, AWS, Bash, MongoDB, Supabase, Git/GitHub, VS Code, Figma

## EXPERIENCE

### AI Software Engineer Intern

May 2026 – Aug 2026

*Cerebras Systems | C++, Python, Assembly, Bash, GTest*

*Toronto, Ontario*

- Cut production fault-triage time **30%** on wafer-scale inference with a fault-targeting layer that **pinpoints faults** to the right compute unit across tensor-parallel, multi-core, and pipeline-parallel execution.
- Extended **hardware fault injection** with three on-demand failure modes: crash, stall, slowdown, wiring them through the full wafer-inference stack from runtime layer to on-device kernel assembly.
- Silenced false on-call pages by redesigning faulty-hardware detection with a two-tier alerting scheme.
- Built developer tooling: a static linter for a custom hardware assembly language and an observability dashboard.

### Front-End Developer

Sep 2023 – Apr 2024

*SolveXchange | HTML, CSS, JavaScript, Flask, SQLAlchemy*

*Online*

- Shipped and maintained responsive frontend interfaces serving **100+** active users for a non-profit platform.
- Built secure authentication and data workflows with **Flask**, **SQLAlchemy**, and **RESTful** APIs.

## PROJECTS

**Quota** — DeltaHacks12 First Place 🏆 | *TypeScript, Next.js, React Flow, Gemini, LangChain, MongoDB*

- Built VS Code extension with **TypeScript** for inline cost annotations, heatmaps, and one-click optimizations.
- Indexed codebases **<3s** for large codebases with **10,000+** lines using **AST parsing** to detect inefficiencies.
- Delivered real-time API cost insights via regex and usage caching, cutting API spend by up to **40%**.
- Created **React Flow** sandbox for budget-aware system planning with **RAG Gemini** chatbot by LangChain.

**Nebula AI** 🏆 🍏 | *React Native, FastAPI, PostgreSQL, Docker, GCP*

- Developed a mobile app that turns handwritten notes into a searchable, **LaTeX**-formatted AI knowledge base.
- Enabled semantic search over **1,536-dim embeddings**, cutting retrieval time by **40%** with IVFFlat indexing.
- Implemented an **OCR+AI pipeline** with Gemini/OpenAI, enabling handwritten text extraction and summaries.
- Designed multi-tenant data access using Supabase Auth, JWTs, and RLS policies across 5 relational tables.
- Containerized backend with **Docker**, deployed to **Google Cloud Run** with automated CI/CD pipelines.

**Personal CRM** 🏆 | *Next.js, FastAPI, PostgreSQL, TailwindCSS*

- Built a robust full-stack CRM to centralize and prioritize personal & professional relationships at scale.
- Architected backend with **20+ REST** endpoints, enabling pagination & dynamic fetching for growing datasets.
- Improved system reliability through caching layers and rate limiting, reducing server load by **30%**.
- Designed a **matrix-based algorithm** to rank contacts, enabling users to focus on high-value relationships.

**Melodie.ai** 🏆 | *PyTorch, NumPy, Python, Music21*

- Developed an **LSTM**-based neural network with **PyTorch** to generate folk melodies, reaching coherent sequences.
- Preprocessed MIDI data by standardizing keys and normalizing sequences for simpler, faster training.
- Designed **probabilistic sampling system** with temp. control and nucleus sampling methods, enabling variation.

## PUBLICATIONS

**Seoh, J. (2025). *Analyzing Literary Factors: A Big Data Approach*. STEM Fellowship Journal.**

Published **academic paper**: "[Literary Factors Analysis](#)" (DOI #: 10.17975/sfj-2025-001).